Henry Silva

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Objective

Proven engineering and product development leader seeking next technology to transition from laboratory to mass market release.

Education

Georgia Institute of Technology | Atlanta, GA *May 2026 – Present*
Bachelor of Science in Electrical Engineering, GPA 3.60 Expected Graduation, May 2028

Purdue University | West Lafayette, IN *August 2019 – May 2023*
Transfer with 45 Credit Hours, GPA 3.78

Experience

Slingshot Product Development Group | Duluth, GA *September 2025 – Present*
New Product Consultant

- Took ownership of embedded systems on a behind-schedule medical IoT device; drove firmware development to resolve RTOS-related issues to get the project back on track and ready for market release.
- Executed complete redesign of critical circuit boards to improve manufacturability and performance from schematic capture through fabrication and assembly drawings.
- Designed and built a custom high-voltage piezo transducer characterization test system enabling repeatable R\&D measurement and automated production QC.
- Established the firm's first centralized version-controlled code infrastructure and drove adoption of Altium Designer as the standard ECAD tool across the engineering team.
- Evaluated potential clients through patent research and discovery calls; interviewed candidates for open embedded systems engineer roles.

Wavelogix | West Lafayette, IN *January 2023 – June 2025*
Co-Founder, Director of Product Development

- Hired and led a new multi-disciplinary engineering team of 6 full-time engineers spanning embedded firmware, PCB hardware, cloud ML, and full-stack web development.
- Designed and deployed a cloud-connected manufacturing system including automated QC test fixtures and firmware programming stations enabling production and validation of 100+ sensor and IoT datalogger units per day.
- Architected and built complete firmware development infrastructure: containerized toolchains with Docker, cloud CI/CD with GitHub Actions, hardware-in-the-loop testing with Raspberry Pi, and LTE FOTA via Azure IoT Hub.
- Drove a complete overhaul of the company's first generation IoT concrete-sensor platform, delivering a market-ready product within 18 months and deploying the MVP across 8 states in summer 2023.
- Transformed team culture from siloed execution to a collaborative, ownership-driven model, reducing management overhead and improving cross-functional velocity.

Wavelogix | West Lafayette, IN *August 2021 – December 2022*
Co-Founder, Product Engineer

- Designed and built the company's first LTE IoT data-logger end-to-end: embedded firmware, PCB design, and full mechanical assembly. Delivered 100 units on schedule for cross-country deployment.
- Managed a web applications consultanting firm to develop prototype cloud infrastructure and user-facing dashboard, defining UX requirements and integration specifications.
- Conducted field testing on active construction sites to validate product reliability and gather direct customer feedback.

Skills

Embedded Systems: STM32, NRF52, ESP32, BLE, LTE, FOTA, Containerized Development
Hardware Design: KiCAD/Altium, Fusion 360/SolidWorks, PCB Prototyping, SMD/THT Soldering, 3D Printing, Injection Molding
Test & Measurement: Analog Discovery 3, WaveForms Scripting, Oscilloscope, Spectrum Analyzer, Logic Analyzer
Software & Infrastructure: Git, GitHub Actions, CI/CD, HIL Testing, Python, MATLAB, Linux
Cloud IoT: Azure IoT Hub, Azure ML Studio, Azure Functions, MongoDB